Stochastic Control Problem: Multistage Stochastic Programming for Portfolio Optimization with Derivatives

MTP Thesis Presentation

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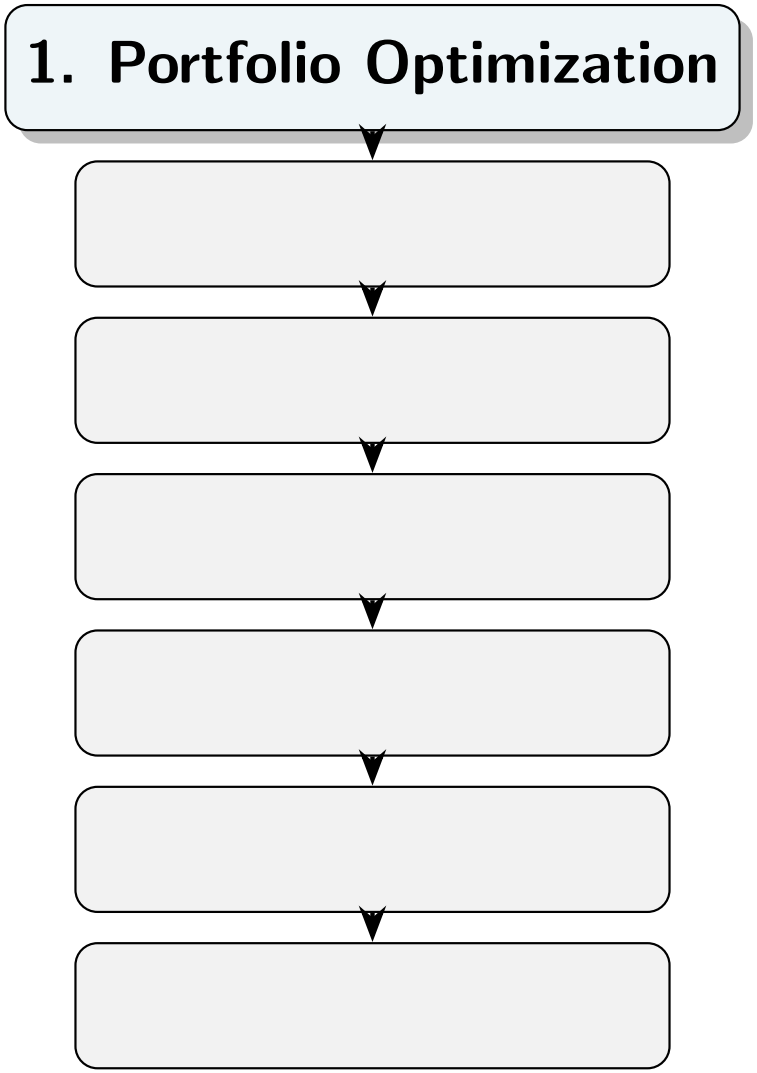
Today's roadmap



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Goal:

To allocate wealth across multiple financial assets to maximize return and/or minimize risk.

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- **Risk:** Uncertainty in returns, often measured by variance or Value-at-Risk.
- **Trade-off:** Higher return typically comes with higher risk.

Static vs Dynamic Portfolio Optimization

Static Optimization:

- Decision is made only once, at the beginning of the investment horizon.
- No adjustment based on how the market evolves.
- Example: One-time allocation of 10 lakh into 3 assets for the next 5 years.

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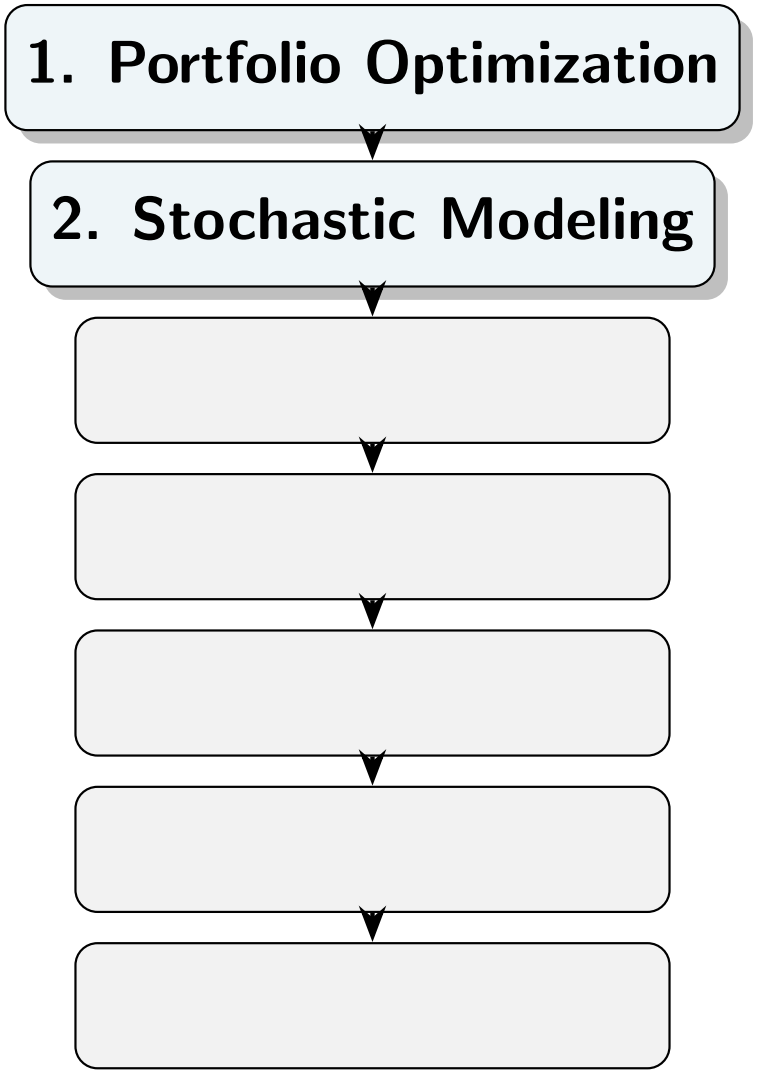
Dynamic Optimization:

- Rebalancing decisions are made at each time step, based on new market info.
- Example: Monthly asset allocation adjustment based on evolving prices and risks.

Today's Journey



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- We model randomness using **stochastic processes**, e.g., Brownian motion.
- This leads us to tools from **Stochastic Calculus** for portfolio modeling.

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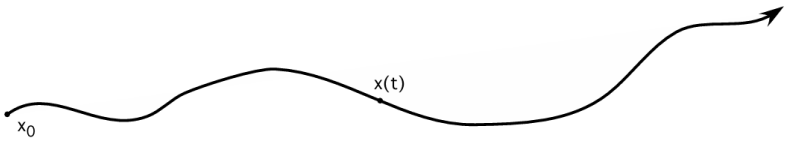

Figure: ODE Trajectory

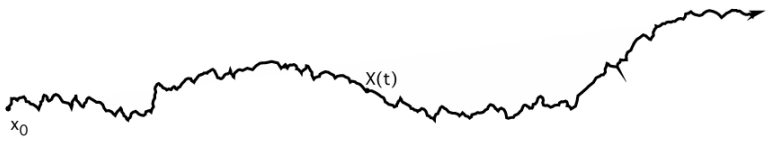

Figure: SDE Sample Path

Modeling Asset Prices using SDEs

In financial markets, asset prices are modeled using a **Stochastic Differential Equation (SDE)**:

$$dP(t) = \mu P(t)\,dt + \sigma P(t)\,dW(t)$$

Where:

- $P(t)$: Asset price at time t

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- σ: Volatility (diffusion coefficient)
- $dW(t)$: Brownian motion increment

Interpretation:

- The first term models **deterministic growth**.
- The second term models **random market shocks**.

Stochastic Wealth Evolution – Multi-Asset Portfolio

Let $X(t)$ be the total portfolio wealth. Using m risky assets and 1 risk-free asset:

$$X(t) = \sum_{i=1}^{m} N_i(t)P_i(t) + N_0(t)P_0(t)$$

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Assuming stock prices follow:

$$dP_i(t) = P_i(t)\left[b_i(t)\,dt + \sum_{j=1}^{m} \sigma_{ij}(t)dW_j(t)\right]$$

$$dP_0(t) = r(t)P_0(t)dt$$

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Substituting and using $\omega_i(t) = N_i(t)P_i(t)$, we get:

Risk Minimization Objective

Goal: Minimize portfolio risk (variance of wealth) over the investment horizon $[0, T]$

Cost Function:

$$\min_{\omega(t)} \mathbb{E}\left[\int_0^T \frac{1}{2}\sum_{i=1}^{m}\sum_{j=1}^{m}\omega_i(t)\,\sigma_{ij}(t)\,\omega_j(t)\,dt\right]$$

Where:

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- $\sigma_{ij}(t)$: covariance between asset i and j
- $\omega(t) \in \mathbb{R}^m$: control vector (decision variable)

This is a **stochastic control problem** with the wealth SDE as the system dynamics.

Hamilton–Jacobi–Bellman (HJB) Equation

Let $V(t,x)$ be the value function (minimum cost-to-go). Then the HJB equation is:

$$\frac{\partial V}{\partial t} + \min_{\omega(t)} \left\{ \frac{1}{2} \sum_{i=1}^{m} \sum_{j=1}^{m} \omega_i(t)\, \sigma_{ij}(t)\, \omega_j(t) + \frac{\partial V}{\partial x} \cdot f(t,x,\omega) \right\} = 0$$

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Where $f(t,x,\omega)$ is the drift from the wealth SDE:

$$f(t,x,\omega) = r(t)x + \sum_{i=1}^{m} (b_i(t) - r(t))\omega_i(t)$$

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Boundary Condition: $V(T,x) = 0$
Goal: Solve for the optimal control $\omega^*(t)$ that minimizes this PDE.

Today's Journey

1. **Portfolio Optimization**
2. **Stochastic Modeling**
3. **Challenges**

Now Entering:

Why Not Solve HJB Directly?

Challenges with the HJB Approach:

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- **Numerical Instability:** PDE solvers become inefficient or unstable in high dimensions.
- **Nonlinearity of Derivatives:** Options introduce payoff discontinuities and path-dependence.

Hence: We switch to a **Multistage Stochastic Programming (MSP)** framework.

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Multistage Stochastic Programming (MSP)

MSP provides a tractable way to handle dynamic decision-making under uncertainty.

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- Objective: maximize expected cost or minimize risk across all scenarios

Why MSP?

- Avoids intractable HJB PDEs
- Adapts to market evolution stage-by-stage

What is a Scenario Tree?

Definition:

- A tree structure to represent market evolution over time.
- Each **node** = market state at time t
- Each **branch** = possible return scenario to next time

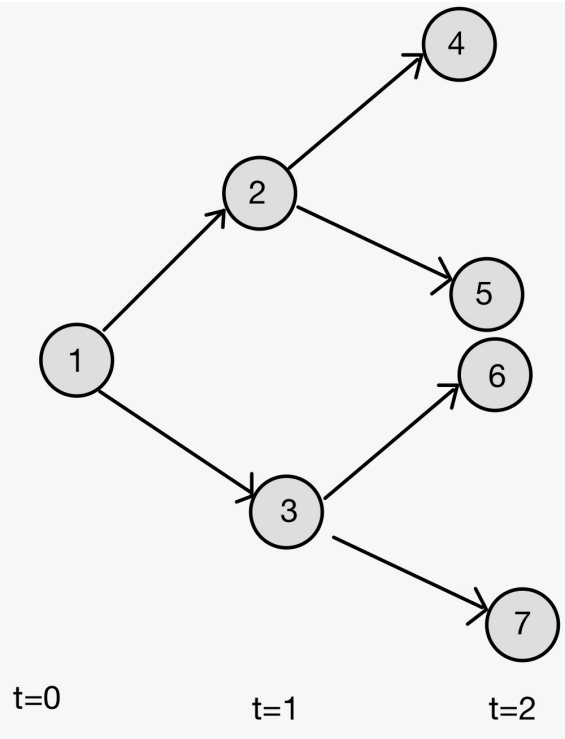


Figure: *

How Do We Construct It?

Steps:

1. **Bootstrap** historical returns to simulate future returns.
2. From each node, sample b return vectors $\rightarrow$ create b children.
3. Repeat for T stages (e.g., 3 time periods).

Result:

- A multi-stage tree with realistic asset return paths.
- Each path from root to leaf = one market evolution scenario.

Ensuring No Arbitrage

Why?
Arbitrage distorts portfolio value and leads to unrealistic pricing.

Check at Each Node:

- Let parent node have return vector r_0
- Children have returns $\{r_1, r_2, \ldots, r_b\}$
- Solve LP:

$$r_0 = \sum_{j=1}^{b} p_j r_j \quad \text{with} \quad \sum p_j = 1,\ p_j \geq 0$$

- If solution exists $\rightarrow$ no arbitrage. Else, resample.

Intuition: Parent return must lie inside the convex hull of its children.

Ensuring No Arbitrage

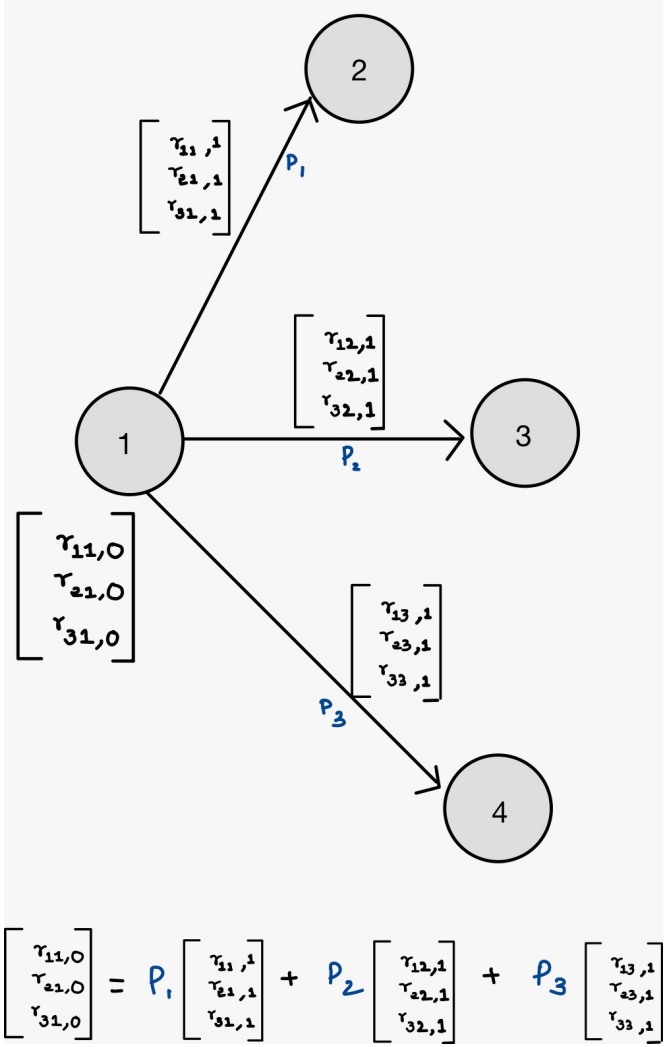


*Figure: Martingale check – Can parent return be expressed as convex combination of children?

Assigning Empirical Probabilities

Assumption: Uniform bootstrapping from historical data.

Recursive Assignment:

$$p(n) = \frac{1}{b} \cdot p(\pi(n)), \quad p(\text{root}) = 1$$

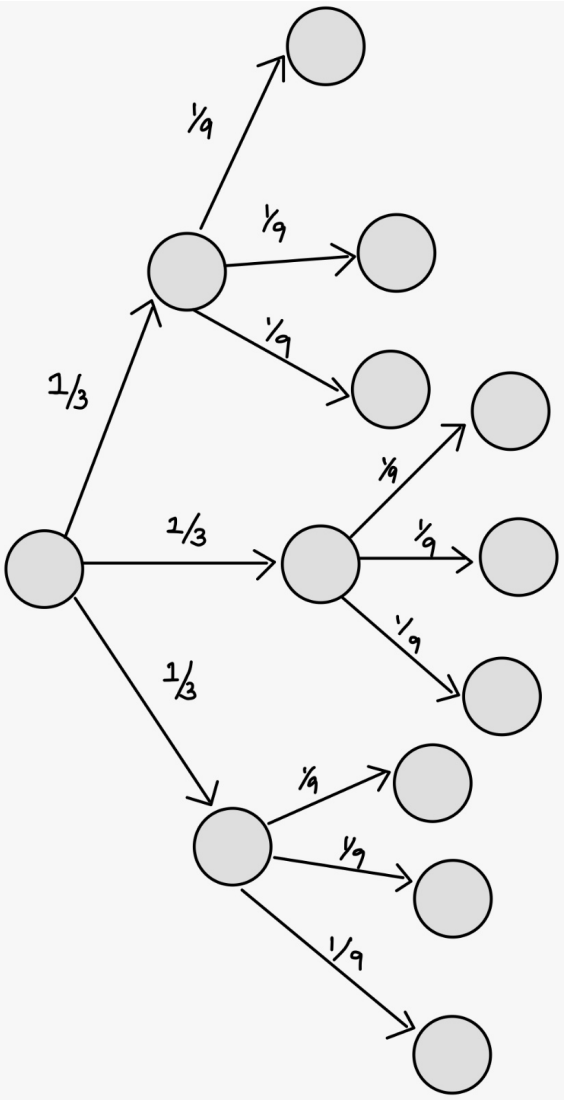
Path Probability:

$$\mathbb{P}(\text{path}) = \left(\frac{1}{b}\right)^T$$

Use in Optimization:

- Compute expected return, risk.
- Acts as real-world probability measure $\mathbb{P}$ before transforming to MEMM-based $\mathbb{Q}$.

Empirical Probabilities in Scenario Tree



*Uniform branching: Each path has equal empirical probability under $\mathbb{P}$

Risk-Neutral Measure via MEMM

Why Change Measure?

- Empirical probabilities $\mathbb{P}$ don't ensure no-arbitrage.
- For pricing derivatives, we need **risk-neutral** probabilities $\mathbb{Q}$.
- Under $\mathbb{Q}$, discounted asset prices must follow a **martingale**.

MEMM Optimization:

$$\mathbb{Q}^* = \arg\min_{q \in \mathcal{M}} \sum_j q_j \log\left(\frac{q_j}{p_j}\right)$$

$$\text{s.t.} \quad \sum_j q_j = 1, \quad \sum_j q_j r_j = r_0, \quad q_j \geq 0$$

Outcome:

- Closest arbitrage-free measure $\mathbb{Q}$ to empirical $\mathbb{P}$.
- Enables **derivative pricing** and valid stochastic optimization.

Intuition Behind MEMM

What is MEMM really doing?

- It finds the **risk-neutral measure** $\mathbb{Q}$ that is:
  - **Martingale-consistent** $\rightarrow$ No arbitrage
  - **Closest to real-world data** $\mathbb{P}$

- Minimizing relative entropy (KL divergence):

$$D_{\mathrm{KL}}(\mathbb{Q}\|\mathbb{P}) = \sum_j q_j \log\left(\frac{q_j}{p_j}\right)$$

- Means: **Don't distort the empirical world too much** just to make it risk-neutral.

Analogy: Like tweaking your forecast just enough to be fair, but still realistic.

Jensen's Inequality – Why KL Divergence ≥ 0

Jensen's Inequality: If f is concave and $\sum_j q_j = 1$, then:

$$f\left(\sum_j q_j x_j\right) \geq \sum_j q_j f(x_j)$$

Apply to: $f(x) = \log(x)$, $x_j = \frac{p_j}{q_j}$

$$\sum_j q_j \log\left(\frac{p_j}{q_j}\right) \leq \log\left(\sum_j p_j\right) = \log(1) = 0$$

Flip the sign:

$$\sum_j q_j \log\left(\frac{q_j}{p_j}\right) \geq 0$$

Why KL Divergence ≥ 0

Conclusion:

- KL divergence is always ≥ 0
- Equals 0 only when $q_j = p_j \; \forall j$

So: Minimizing KL pulls q as close to p as possible.

In MEMM:

- You're forced to adjust p (to be risk-neutral)
- But you adjust it **minimally** — just enough to eliminate arbitrage

European Call and Put Option Payoffs

European Options: Can be exercised only at maturity time T

Call Option:

- Right to buy the asset at strike price K
- Payoff at node $n \in T$:

$$\phi_n^{\text{call}} = \max(S_n - K, 0)$$

Put Option:

- Right to sell the asset at strike price K
- Payoff at node $n \in T$:

$$\phi_n^{\text{put}} = \max(K - S_n, 0)$$

Used in optimization as terminal cashflows at leaf nodes.

Stock Price Evolution on Scenario Tree

At each node n:

$$S_n = S_{\text{parent}(n)} \cdot (1 + r_n)$$

Base Case: S_0 is the initial stock price

Recursive Formula:

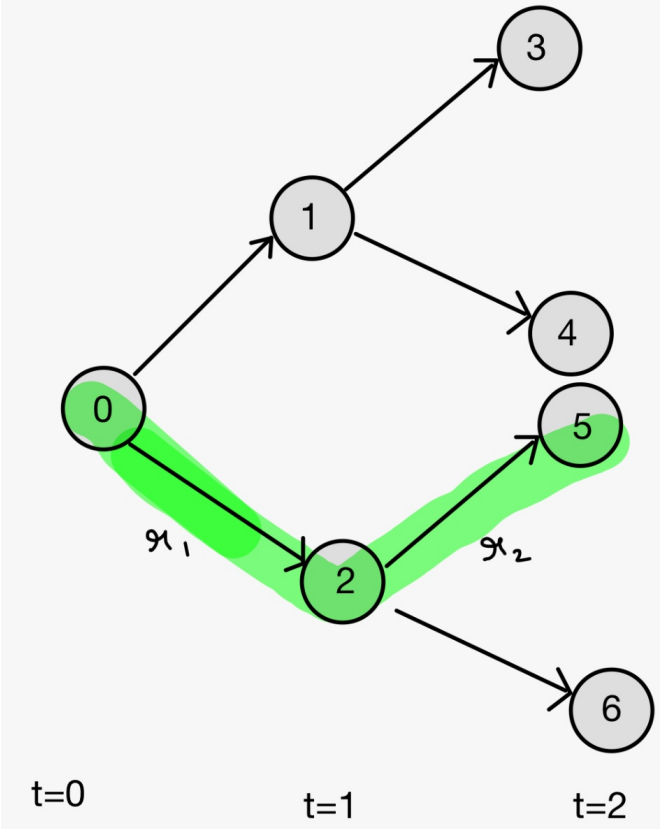
$$S_n = S_0 \cdot \prod_{k \in \text{path}(n)} (1 + r_k)$$

Where:

- r_n: asset return at node n
- $\text{path}(n)$: sequence of nodes from root to n

Terminal prices S_n are used to compute option payoffs ϕ_n.

Stock price Representation



$$S_5 = S_2(1+r_2)$$

$$S_2 = S_0(1+r_0)$$

Option Pricing via Backward Recursion

Goal: Compute fair price π_0 of the option at the root node

Step 1 – Terminal Payoff:

$$\pi_n = \phi_n = \max(S_n - K, 0) \quad \text{for all terminal nodes } n \in T$$

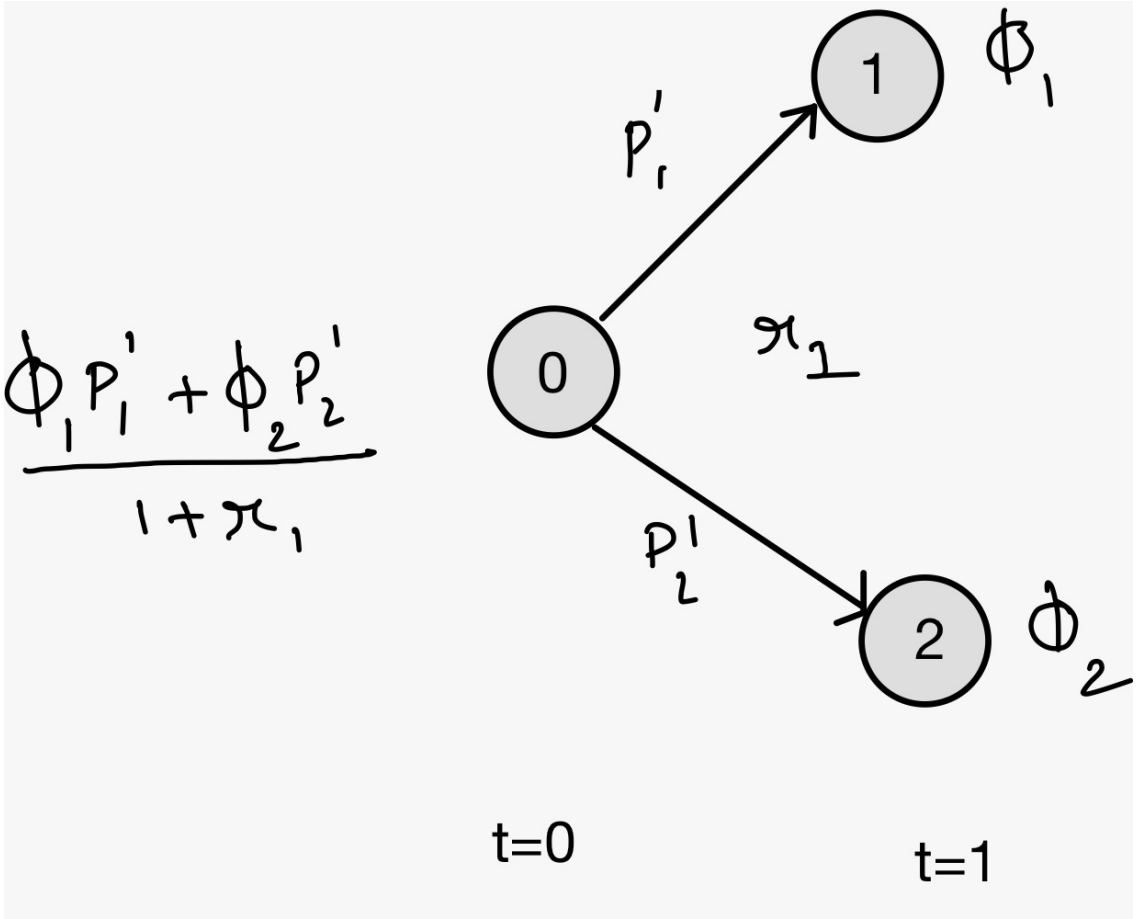
Step 2 – Recursive Valuation:

$$\pi_n = \frac{1}{1 + r_n} \sum_{j \in \text{child}(n)} q_{nj} \cdot \pi_j \quad \text{for } n \in T - 1, \ldots, 0$$

Where:

- π_n: option value at node n
- q_{nj}: MEMM probability from parent n to child j
- r_n: interest rate at node n

example



Wealth Propagation – Node-wise Recursion

For any non-root node $n \in \mathcal{N} \setminus \{0\}$, wealth is given by:

$$W(n) = \sum_{i=1}^{d} x_i(\pi(n)) \cdot W(\pi(n)) \cdot R_i(n) + C(n)$$

Where:

- $x_i(\pi(n))$: fraction of wealth invested in risky asset i at parent node $\pi(n)$
- $W(\pi(n))$: wealth at the parent node $\pi(n)$
- $R_i(n)$: return of asset i from parent to node n
- $C(n)$: total cashflow from options arriving at node n

This provides a recursive formula where each asset investment is a proportion of prior node wealth.

How Options Impact Wealth Flow

Cashflow from options at node n:

$$C(n) = \sum_{j=1}^{J} v_{j-1}(\pi(n)) \cdot \begin{cases} \text{Payoff}_j(n), & \text{if } t(n) = \tau_j \\ V_j(n), & \text{otherwise} \end{cases}$$

Interpretation:

- $v_{j-1}(\pi(n))$: number of units of option j bought at parent node
- At maturity: option gives payoff
- Otherwise: contributes mark-to-market value

So total wealth $W(n)$ includes:

$$\text{Asset value} + \text{Option cashflow}$$

Option Holdings and Units Purchased

At each parent node $\pi(n)$, suppose you allocate a fraction $y_j(\pi(n))$ of total wealth to option j.

Then, the number of units purchased is:

$$v_j(\pi(n)) = \frac{y_j(\pi(n)) \cdot W(\pi(n))}{V_j(\pi(n))}$$

Where:

- $y_j(\pi(n))$: fraction of wealth allocated to option j
- $W(\pi(n))$: total wealth at parent node
- $V_j(\pi(n))$: price of option j at node $\pi(n)$

So, option allocation decisions are proportional and wealth-based.

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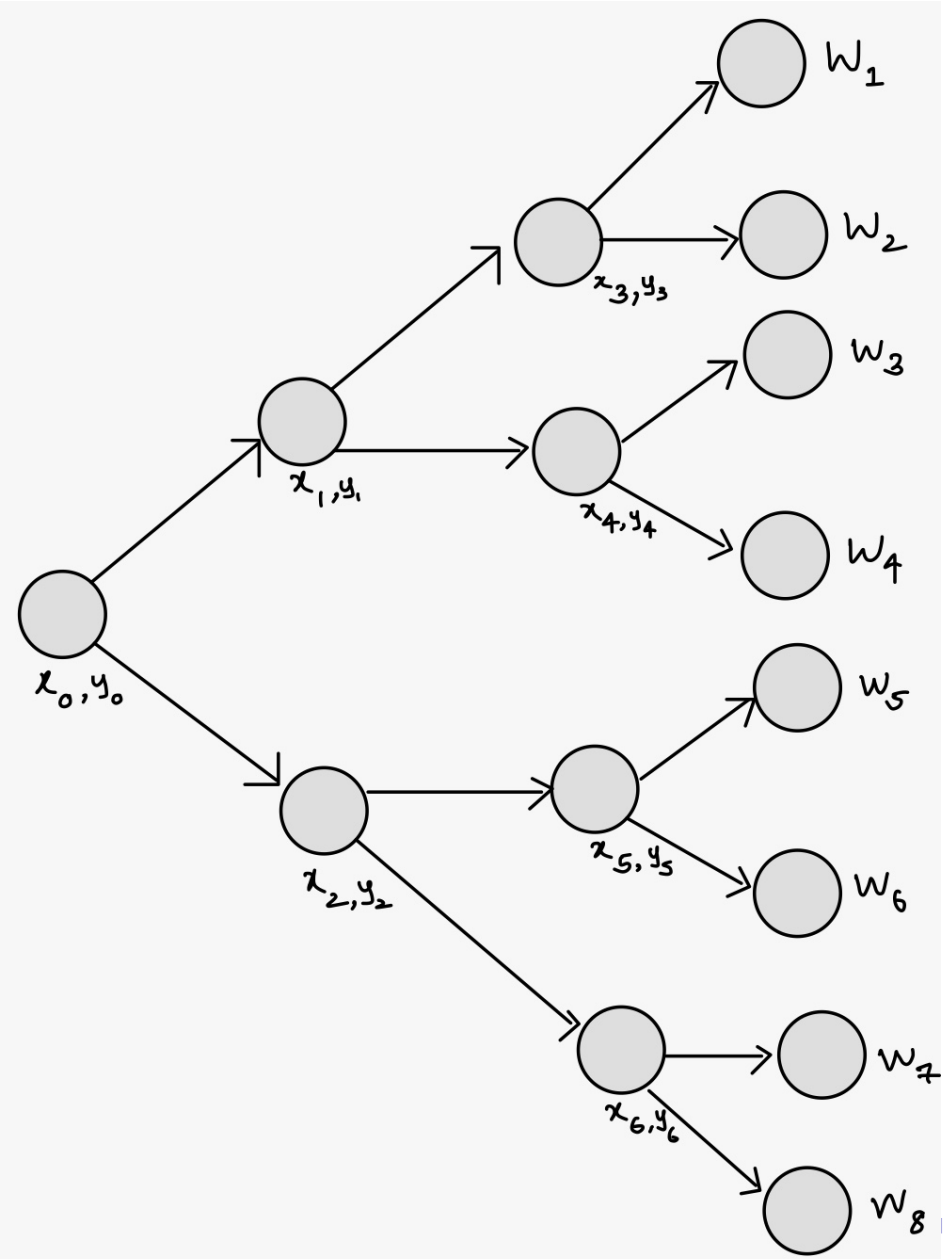
5. Optimization

Scenario Tree Representation



MSP Objective – Maximize Expected Terminal Wealth

Goal: Maximize the expected value of terminal wealth across all scenarios

$$\max \mathbb{E}[W_T]$$

Where:

- W_T: terminal wealth (random variable) at time horizon T
- Expectation taken over all possible scenario paths

This is a multistage problem:

- Wealth depends on investment and option decisions at every stage
- Optimization must plan decisions at all nodes to influence W_T

Why Max-Expectation Isn't Enough: Enter CVaR

Problem: Maximizing $\mathbb{E}[W]$ alone ignores downside risk

Observation:

- Two portfolios can have same expected return...
- But very different outcomes in bad scenarios

Two Portfolios with Equal Expectation:

Scenario	Portfolio A	Portfolio B
Best (30%)	1200	1900
Medium (40%)	1000	1000
Worst (30%)	900	200

$$\mathbb{E}[W_A] = \mathbb{E}[W_B] = 1020$$

But Look at the Downside:

- A's worst-case loss: mild $\rightarrow$ 900
- B's worst-case loss: extreme $\rightarrow$ 200

CVaR$_{30\%}$: $\mathrm{CVaR}_{30\%}(A) = 900, \quad \mathrm{CVaR}_{30\%}(B) = 200$

Insight:

- Portfolio A is more stable and balanced
- Portfolio B chases return but exposes you to disaster

CVaR helps the optimizer avoid these risky tails!

Solution

Solution: Add CVaR penalty

$$\mathrm{CVaR}_\alpha(W) = \mathbb{E}[W \mid W \leq \mathrm{VaR}_\alpha]$$

Why CVaR?

- Captures expected loss in worst α
- Convex, so optimization stays tractable
- Gives risk-aware control over portfolio tail losses

Next: How CVaR is added to the optimization

CVaR-Based Risk-Averse Optimization

CVaR Definition:

$$\mathrm{CVaR}_\alpha(W) = \frac{1}{\alpha} \sum_{n \in \mathcal{L}_\alpha} \frac{1}{|\mathcal{L}|} W(n)$$

- $\mathcal{L}_\alpha \subset \mathcal{L}$: worst $\alpha \cdot |\mathcal{L}|$ terminal nodes
- Minimizing CVaR focuses on tail risk (bad outcomes)

Risk-Averse Objective:

$$\min_{x,y} \quad \mathrm{CVaR}_\alpha(W)$$

- Subject to same feasibility and allocation constraints
- Encourages stable and conservative strategies

Risk-Return Tradeoff Formulation

Combined Objective:

$$\max_{x,y} \quad (1-\lambda)\cdot\mathbb{E}[W] - \lambda\cdot\mathrm{CVaR}_\alpha(W) \quad \text{for } \lambda \in [0,1]$$

Tunable Risk Preferences:

- $\lambda = 0$: Pure return maximization (risk-neutral)
- $\lambda = 1$: Pure CVaR minimization (fully risk-averse)
- $0 < \lambda < 1$: Balanced trade-off between return and risk

This gives a unified and flexible control over portfolio behavior.

Today's Journey

1. Portfolio Optimization
2. Stochastic Modeling
3. Challenges
4. MSP Framework
5. Optimization
6. Implementation and Results

Implementation Framework

Dataset:

- Monthly historical prices from Jan 2004 to Dec 2021.
- Assets: S&P500, GOOG, AAPL, MSFT, and 1 risk-free (T-bill).
- Returns computed as:

$$r_{i,t} = \frac{P_i(t) - P_i(t-1)}{P_i(t-1)}$$

Preprocessing:

- Cleaned missing values, aligned time series.
- Resampled to consistent monthly intervals.

Scenario Tree Construction

- **Tree Parameters:**
  - Depth $T = 3$, Branching factor $b = 10$
  - Total nodes: $1 + 10 + 100 + 1000 = 1111$
- **Return Sampling:**
  - Bootstrapped from historical monthly return vectors.
  - Simulates realistic market movements.
- **Arbitrage Filtering:**
  - LP check to ensure no-arbitrage at each branching.

Option Pricing using MEMM

Objective: Compute fair prices of European options using arbitrage-free scenario tree.

Step 1: Risk-Neutral Probability Calculation

- Applied **Minimal Entropy Martingale Measure (MEMM)** at each branching.
- Transformed real-world probabilities into risk-neutral ones.
- Solved constrained optimization problem at each parent node.

Step 2: Backward Option Valuation

- Option payoff computed at maturity nodes.
- Backward recursion applied using MEMM probabilities.
- Prices computed for 8 options (1 call + 1 put on each of 4 assets).

Result: Option prices available at root node — used for optimization and comparison.

Option Prices at Root Node (Stage 0)

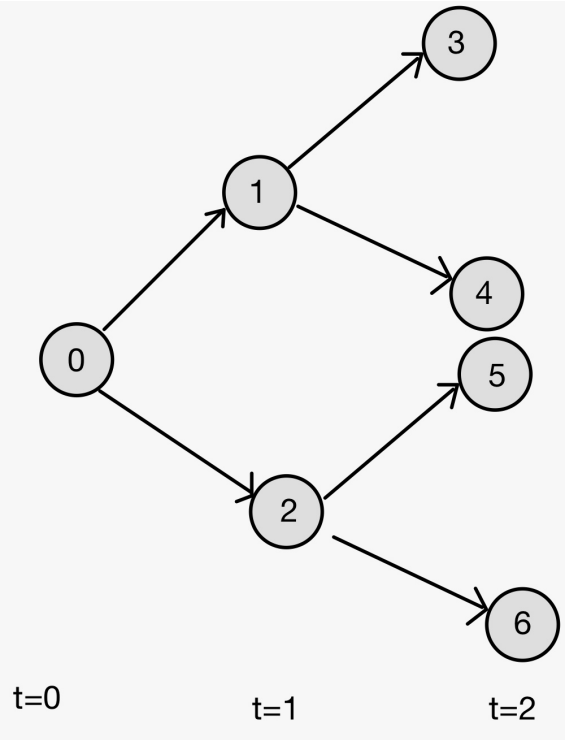
Setup:

- European call and put options priced via backward recursion.
- Strike set at 105% of each asset's initial value.
- All options mature at Stage 3.

Type	Asset	Strike	Value at Root
Call	SP500Close	2765.89	82.53
Put	SP500Close	2765.89	45.62
Call	GOOGClose	3015.77	90.01
Put	GOOGClose	3015.77	51.12
Call	AAPLClose	150.12	12.34
Put	AAPLClose	150.12	6.78
Call	MSFTClose	210.23	17.56
Put	MSFTClose	210.23	9.45

Option values reflect MEMM-based pricing using arbitrage-free tree.

3-Stage Binary Tree Structure



- **Stage 0:** Node 0 (Root)
- **Stage 1:** Nodes 1, 2
- **Stage 2:** Nodes 3, 4, 5, 6 (Leaves)

Flattened Tree Representation

Node ID	Stage	Parent	Returns [r1, r2]
0	0	-1	None
1	1	0	[0.05, -0.02]
2	1	0	[0.03, 0.01]
3	2	1	[0.01, 0.01]
4	2	1	[-0.02, 0.03]
5	2	2	[0.02, -0.01]
6	2	2	[0.04, 0.00]

Allocations and Initial Wealth

- **Decision Nodes:** 0, 1, 2
- **Optimizer Output (X):**

$$X_0 = [0.6,\ 0.4]$$
$$X_1 = [0.7,\ 0.3]$$
$$X_2 = [0.5,\ 0.5]$$

- **Initial Wealth:** $W_0 = 10{,}000$

Wealth Recursion (Forward)

Node	Wealth Formula	Result
1	$10{,}000 \times (1 + 0.6 \cdot 0.05 + 0.4 \cdot (-0.02))$	10,240
2	$10{,}000 \times (1 + 0.6 \cdot 0.03 + 0.4 \cdot 0.01)$	10,220
3	$10{,}240 \times (1 + 0.7 \cdot 0.01 + 0.3 \cdot 0.01)$	10,342.4
4	$10{,}240 \times (1 + 0.7 \cdot (-0.02) + 0.3 \cdot 0.03)$	10,144.64
5	$10{,}220 \times (1 + 0.5 \cdot 0.02 + 0.5 \cdot (-0.01))$	10,276.21
6	$10{,}220 \times (1 + 0.5 \cdot 0.04 + 0.5 \cdot 0.00)$	10,424.4

How Optimization Works (with Example)

Step 1: Start with Initial Allocation X

- Optimizer begins with some guess:

$$X_0 = [0.6,\ 0.4], \quad X_1 = [0.7,\ 0.3], \quad X_2 = [0.5,\ 0.5]$$

Step 2: Compute Wealth Recursively

- Using these X_i, forward recursion is used to compute:

$$W[i] = W[\text{parent}] \cdot (1 + x_{\text{parent}} \cdot r_i)$$

- Wealths obtained at all nodes (see next slide).

Step 3: Evaluate Objective Function

- Terminal node wealths W_T are used to compute:

$$\max_X \left((1-\lambda) \cdot \mathbb{E}[W_T] - \lambda \cdot \text{CVaR}_\alpha(W_T)\right)$$

- Or pure expected wealth if $\lambda = 0$

Step 4: Update and Iterate

- SLSQP updates X to improve the objective.

Why Flattening is Crucial

- Enables easy interfacing with `SciPy`'s `minimize` function.
- Allows constraints to be applied across stages and decision nodes.
- Speeds up computation: linear-time access to stage, parent, returns.
- Allows simulation of thousands of paths and strategies efficiently.

Expected Wealth Maximization

- **With Options:** The expected terminal wealth was approximately \$16,583.16.
- **Without Options:** The expected terminal wealth was \$15,443.03

With CVaR Optimization ($\alpha = 0.95$)

- **With Options:** The strategy produced a mean terminal wealth of \$10,682.38.
- **Without Options:** The final CVaR was \$10,423.32

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Conclusion

- Scenario tree model enables dynamic portfolio optimization across time.
- Flattened tree structure allows efficient wealth recursion and vectorized optimization.
- MEMM ensures arbitrage-free option pricing within the tree.
- Results demonstrate:
  - **Higher expected returns** with options (Expected Wealth)
  - **Improved downside protection** (CVaR objective)
- Options play a vital role in both return enhancement and risk mitigation.

A flexible, robust framework for multi-stage, risk-aware portfolio optimization.

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